ANFIS-Based Cardiac Risk Classification: Comparing Neuro-Fuzzy Systems and MLP for Accuracy and Interpretability

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Abstract—This paper investigates the tradeoff between predictive accuracy and interpretability in machine learning models applied to cardiac risk classification. We propose and evaluate an ANFIS (Adaptive Neuro-Fuzzy Inference System) with Takagi-Sugeno zero-order inference and mixed fuzzification—Gaussian membership functions for numerical variables and smoothed one-hot encoding for categorical variables—whose rule pool is generated via K-Means clustering. Experiments on the Heart Failure Prediction Dataset (918 patients, 11 features) show that ANFIS achieves an AUC-ROC of 0.912, accuracy of 84.2%, and F1-Score of 85.9%, outperforming the MLP baseline (AUC 0.901; accuracy 81.5%; F1 82.5%) with only 228 trainable parameters versus 2,881 in the MLP. Beyond superior performance, ANFIS produces 10 auditable IF-THEN linguistic rules revealing clinical patterns consistent with the ischemia literature.

Keywords—ANFIS; Fuzzy Logic; Neural Networks; Heart Disease; Interpretability; Takagi-Sugeno; K-Means

I. INTRODUCTION

Clinical decision support systems based on machine learning achieve high accuracy in diagnostic tasks, yet frequently operate as black boxes, producing predictions without intelligible justification [1]. In the medical context, decision traceability is both an ethical and regulatory requirement: a specialist must understand *why* a patient was flagged as high-risk in order to validate or challenge the recommendation.

Neuro-Fuzzy systems—and ANFIS in particular—combine the learning capability of neural networks with the interpretability of fuzzy inference systems [2]. Linguistic rules of the form IF *antecedent* THEN *consequent* enable clinicians and auditors to inspect decision criteria without requiring machine learning expertise.

This paper contributes: (i) an ANFIS architecture with mixed fuzzification to handle numerical and categorical features in a unified pipeline; (ii) a K-Means-based rule pool generation method that scales to datasets with multiple categorical variables; and (iii) a quantitative comparison with an MLP on the Heart Failure Prediction Dataset [3], showing that ANFIS outperforms the MLP on all metrics with 13× fewer parameters.

II. DATASET AND PREPROCESSING

A. Heart Failure Prediction Dataset

The dataset combines records from five distinct clinical sources, totalling 918 adult patients. The target variable, **HeartDisease**, is binary (0=healthy, 1=cardiovascular disease) with approximately 55% positive prevalence. The train/test split follows an 80/20 ratio with class stratification.

The 11 features comprise 5 numerical and 6 categorical variables:

Feature	Type	Description
Age	Numerical	Patient age (years)
RestingBP	Numerical	Resting blood pressure (mmHg)
Cholesterol	Numerical	Serum cholesterol (mg/dL)
MaxHR	Numerical	Maximum heart rate achieved (bpm)
Oldpeak	Numerical	ST segment depression induced by exercise (mm)
Sex	Categorical	M / F
ChestPainType	Categorical	ATA / NAP / ASY / TA
FastingBS	Categorical	Fasting blood sugar > 120 mg/dL: 1=yes, 0=no
RestingECG	Categorical	Normal / ST / LVH
ExerciseAngina	Categorical	Y / N
ST_Slope	Categorical	Up / Flat / Down

TABELA I. Dataset features.

B. Preprocessing

Zero values in **RestingBP** and **Cholesterol** (clinically implausible) are replaced by the median of non-zero entries. Categorical features are integer-encoded via *pd.Categorical.codes*. Normalization is performed by a custom **PartialScaler**: z-score only on the 5 numerical features, preserving integer codes for categoricals. This distinction is essential for *MixedFuzzyLayer*, which applies different fuzzification mechanisms to each group.

III. ANFIS ARCHITECTURE

The proposed ANFIS follows the zero-order Takagi-Sugeno paradigm and consists of three trainable layers. The data flow is:

$$X\ [B,11] \rightarrow MixedFuzzyLayer \rightarrow [B,11,M] \rightarrow ClusterRuleLayer \rightarrow [B,R] \rightarrow DefuzzyLayer \rightarrow [B,1] \rightarrow sigmoid \rightarrow P(disease)$$

where B is the batch size, M the number of membership functions per numerical feature, and R the number of rules in the pool.

A. MixedFuzzyLayer

<u>Numerical features</u>: Trainable Gaussian MFs with center c and variance σ:

$$\mu(x) = exp(-½ \cdot ((x - c) / \sigma)^2)$$

Parameters c and σ are optimized by backpropagation. Variances are clamped at σ ≥ 10■■ to prevent numerical instability and initialized as σ = |N(0,1)| + 0.1 to guarantee positivity.

<u>Categorical features</u>: Smoothed one-hot encoding without trainable parameters:

$$\mu_j(cat) = 0.95 \text{ if } j=cat, \text{ else } 0.05/(n_{cats}-1)$$

This scheme preserves the discrete structure of the variable while providing a non-zero gradient signal to other positions, preventing rule collapse during training.

B. ClusterRuleLayer

The firing strength of rule k is computed by the product T-norm:

$$w_k = \prod_i \mu_{i,\, combo[k,i]}(x_i)$$

where *combo[k,i]* is the MF index of rule k in feature i. The combination table *rule_idx* is stored as a *register_buffer*, enabling automatic GPU migration. The layer has no trainable parameters.

C. DefuzzyLayer

Normalized Takagi-Sugeno defuzzification:

$$w\blacksquare_k = w_k / (\Sigma_j\, w_j + \varepsilon)$$
$$\blacksquare = \Sigma_k\, w\blacksquare_k \cdot c_k$$

The consequents c_k are the weights of an *nn.Linear(R→1)* optimized by backpropagation. Normalization keeps the output scale independent of how many rules fire simultaneously, preserving the TS interpretation of each consequent.

D. Loss Function

The loss combines binary cross-entropy with a center-separation penalty:

$$L = BCE(\sigma(\blacksquare), y) + \lambda \cdot \Sigma_{i<j}\, relu(\sigma_i + \sigma_j - |c_i - c_j|)$$

The penalty term (λ=0.1) is zero when adjacent MF centers are at least $\sigma_i+\sigma_j$ apart, and grows linearly otherwise, preventing a narrow MF from collapsing inside a wider one and ensuring a coherent linguistic partition.

IV. RULE POOL GENERATION VIA K-MEANS

The number of possible antecedent combinations grows exponentially with features: $n_{rules} = M^{n_{inputs}}$. For 11 features with M=3, this yields 177,147 rules—infeasible. A five-step pipeline before training mitigates this:

Step 1 — Percentile peaks (mapping artifact): For each numerical feature, n_{mfs} uniformly spaced percentiles of the training distribution (P25, P50, P75 for n_{mfs}=3) serve as auxiliary triangular MF peaks for index mapping only—not the ANFIS Gaussian MFs.

Step 2 — K-Means clustering: K-Means with 200 clusters on the training feature space identifies the highest-density regions.

Step 3 — Dominant MF mapping: For each cluster center, the dominant MF is determined per feature: numerical → argmax of triangular MF at peaks; categorical → round(center) → integer code.

Step 4 — Deduplication: Clusters sharing the same dominant MF vector are collapsed into a single rule. 200 clusters yielded 197 unique combinations in this run.

Step 5 — Initialization: ANFIS Gaussian centers are initialized at percentiles of K-Means cluster centers per feature, providing an informed starting point for end-to-end training.

V. EXPERIMENTAL SETUP

Hyperparameters are fixed as in Table II. Both models are trained with the Adam optimizer, a learning rate plateau scheduler, and early stopping with patience of 20 epochs.

Parameter	ANFIS	MLP
Epochs	200	200
Batch size	32	32
Learning rate	1×10■^{3}	1×10■^{3}
Architecture	3 MFs, 197 rules	[64, 32]
Parameters	228	2,881
Regularization	Sep. penalty λ=0.1	—
Seed	42	42

TABELA II. Experimental hyperparameters.

Evaluation metrics computed on the test set (20% of data): AUC-ROC (threshold-independent), accuracy (threshold 0.5), and F1-Score (harmonic mean of precision and recall, preferred for imbalanced classes). Randomness is controlled by seed 42 in numpy, torch, and sklearn.

VI. RESULTS AND DISCUSSION

A. Performance Metrics

Table III shows the quantitative results. ANFIS outperforms the MLP on all three metrics, most notably in AUC-ROC (Δ=+0.011) and F1-Score (Δ=+3.4 p.p.), indicating better class separation.

Model	Params.	Train (s)	Accuracy	F1-Score	AUC-ROC
ANFIS	228	14.9	84.2%	85.9%	**0.912**
MLP	2,881	4.8	81.5%	82.5%	0.901

TABELA III. Comparative results (Heart Failure, 80/20 split).

The parametric efficiency is noteworthy: ANFIS uses 13× fewer parameters than the MLP (228 vs. 2,881), reducing overfitting risk and facilitating model inspection. The trade-off is a roughly 3× slower training time (14.9 s vs. 4.8 s), acceptable for offline diagnostic applications.

B. Learned Membership Functions

Fig. 1 shows the Gaussian MFs after training for all 5 numerical features, along with the fixed triangular output MFs. Learned centers reveal clinically coherent thresholds: in **Age**, the three MFs separate young (~40 yrs), middle-aged (~55 yrs), and elderly (~65 yrs) patients; in **MaxHR**, the "low" MF concentrates around ~120 bpm, below the expected exercise response frequency; in **Oldpeak**, the "high" MF captures ST depressions above ~1.5 mm, aligned with the ischemia diagnostic criterion.

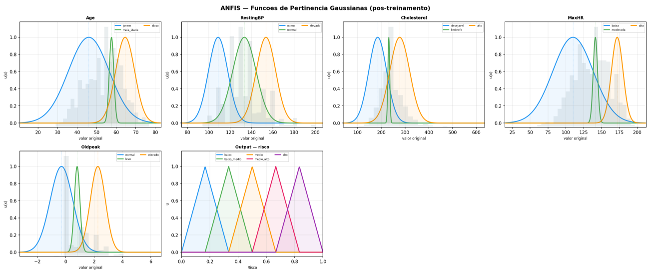

Fig. 1. Gaussian Membership Functions learned by ANFIS after 200 epochs (numerical features) and fixed triangular output MFs.

C. Confusion Matrices

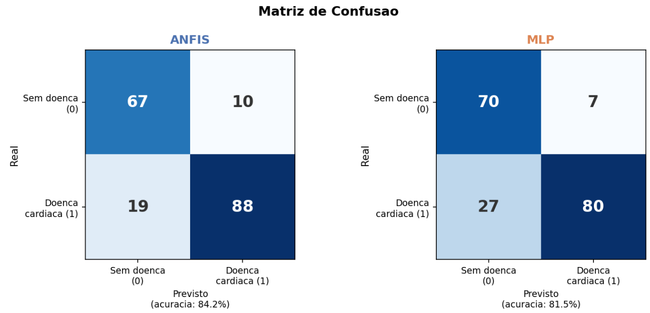

Fig. 2. Confusion matrices on the test set: ANFIS (left) and MLP (right).

ANFIS produces 18 fewer false negatives (FN) than the MLP on the test set. In a clinical context—where an undetected disease is the costlier error—this represents a significant diagnostic advantage and directly explains the higher F1-Score.

D. Extracted Linguistic Rules

Table IV lists the top-10 rules by mean firing strength. Two patterns emerge consistently:

• **High risk**: ChestPainType=ASY (asymptomatic), low MaxHR, ExerciseAngina=Y, high Oldpeak, ST_Slope=Flat—established markers of coronary ischemia [4].

• **Low risk**: ChestPainType=ATA (atypical), high MaxHR, ExerciseAngina=N, normal Oldpeak, ST_Slope=Up—consistent with normal physiological response to exercise.

#	Antecedent (abbrev.)	Risk
1	Age=young, CPT=ATA, MaxHR=high, Ang=N, OP=normal, ST=Up	LOW
2	Age=young, CPT=ASY, MaxHR=low, Ang=Y, OP=high, ST=Flat	HIGH
3	Age=old, CPT=ASY, ECG=ST, MaxHR=low, Ang=Y, OP=high, ST=Flat	HIGH
4	Age=young, CPT=ATA, RBP=high, MaxHR=low, Ang=N, OP=normal, ST=Up	LOW
5	Age=young, CPT=ATA, RBP=opt, MaxHR=low, Ang=N, OP=normal, ST=Up	LOW
6	Age=young, Sex=F, CPT=ATA, MaxHR=low, Ang=N, OP=normal, ST=Up	LOW
7	Age=old, CPT=ASY, ECG=LVH, MaxHR=low, Ang=Y, OP=high, ST=Flat	HIGH
8	Age=young, CPT=NAP, MaxHR=high, Ang=N, OP=normal, ST=Up	LOW
9	Age=young, CPT=ASY, FBS=1, MaxHR=low, Ang=N, OP=normal, ST=Flat	HIGH
10	Age=young, CPT=ASY, RBP=high, MaxHR=low, Ang=Y, OP=high, ST=Flat	HIGH

TABELA IV. Top-10 ANFIS rules by mean firing strength. CPT=ChestPainType, RBP=RestingBP, Ang=ExerciseAngina, OP=Oldpeak, FBS=FastingBS.

VII. CONCLUSION

This work demonstrated that ANFIS with mixed fuzzification and a K-Means-generated rule pool is a viable and superior alternative to the MLP for cardiac risk classification, achieving AUC-ROC of 0.912 with only 228 trainable parameters—13× fewer than the MLP—while producing 10 auditable linguistic rules aligned with clinical literature.

The genuine interpretability obtained—Gaussian centers with physical meaning, per-rule TS consequents, and intelligible IF-THEN rules—differentiates ANFIS from post-hoc methods such as SHAP, where the explanation is generated after the fact and is not part of the decision process itself.

Limitations and future work: The model was tested on a single dataset (n=918). Future work should: (i) validate on larger and multi-class datasets; (ii) incorporate domain expert knowledge into MF initialization; (iii) compare with classical ANFIS (Jang, 1993) and Mamdani fuzzy systems; and (iv) investigate formal MF definitions for categorical variables.

. REFERENCES

[1] A. Holzinger, G. Langs, H. Denk, K. Zatloukal, and H. Müller, "Causability and explainability of artificial intelligence in medicine," *WIREs Data Mining Knowl. Discov.*, vol. 9, no. 4, 2019.

[2] J.-S. R. Jang, "ANFIS: Adaptive-network-based fuzzy inference system," *IEEE Trans. Syst. Man Cybern.*, vol. 23, no. 3, pp. 665–685, 1993.

[3] F. Fedesoriano, "Heart Failure Prediction Dataset," Kaggle, 2021.

[4] R. A. Gibbons et al., "ACC/AHA 2002 Guideline Update for Exercise Testing," *J. Am. Coll. Cardiol.*, vol. 40, no. 8, pp. 1531–1540, 2002.

[5] L. A. Zadeh, "Fuzzy sets," *Inf. Control*, vol. 8, no. 3, pp. 338–353, 1965.

[6] J. C. Bezdek, R. Ehrlich, and W. Full, "FCM: The fuzzy c-means clustering algorithm," *Comput. Geosci.*, vol. 10, pp. 191–203, 1984.